

Name: brightlearn-de-exam-mpho-matebesi
Course: Data Engineering
Project Name: BrightLearn Data Warehouse Build
Project Code: BL-DE-EXAM-2026-07

BrightMart: Data Architecture Plan

1. Where the data comes from

Our point-of-sale (POS) systems dump all standard sales transactions directly into a single flat CSV file.

- **The issue:** The file is pretty messy. It throws customer details, product lines, and store locations together into a giant sheet with zero sorting, and there are plenty of formatting errors.
- **What we will do:** We are going to save a direct copy of this original, untouched CSV file in a locked backup folder. That way, we always have a paper trail if something breaks later on or if we need to audit the records.

2. The staging area

Before building anything complex, we need a workspace to clean up the mess. We will use SQL Server to create some basic temporary tables that match our CSV layout exactly.

Why do this?

- To spot duplicate records, missing blocks, or bad numbers.
- To fix broken dates, drop accidental double entries, and fill in blank spots.
- To keep a clear trail showing exactly how the data changed from the raw file to the final report.

Note: We will use basic SSIS Flat File components to move the files into these temp tables.

3. Sorting out the tables (The Star Schema)

We will organize our clean database using a straightforward center-and-branch layout (a star schema).

- **The core table (FactTransactions):** This is the master list. It records the actual numbers for each sale—how much money came in, what discounts were given, the date, and the specific IDs for the store, customer, product, and how they paid.
- **The context tables (Dimensions):** These are smaller lookup lists that surround our master table. They hold the descriptive details for our Customers, Products, Calendars, Payment Methods, and Store Locations.

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4. Cleaning and moving things over (ETL)

This is where we actually run our processing rules using basic SQL scripts and standard SSIS tools (like Lookups and Conditional Splits).

- **Pulling it out:** We grab the raw data sitting in our temporary staging tables.
- **Fixing it up:** We check things against our basic business rules. For instance, we will cap discounts so they cannot exceed the total item price, turn negative totals into refunds, force all dates into a standard format, and clean up payment names so they read cleanly as Cash, Debit, Credit, EFT, or Store Credit.
- **Loading it up:** We push the tidy, verified data straight into our permanent warehouse tables.

5. Final storage layer

When we write to the main warehouse, we load our lookup lists first. Each unique customer or product gets its own basic tracking key (a surrogate key) to keep things consistent across the system. After that, the main sales list logs transactions using those exact keys.

Doing it this way makes running fast reports a lot easier when management wants to see store performance, total sales by city, or loyalty tier breakdowns.

6. Making the reports

For the final output, we will plug the database into Power BI to create simple dashboards. We will also run raw SQL checks inside SSMS just to make sure the dashboard numbers match the actual warehouse counts.

What the team will see:

- Which stores are performing the best.
- Total revenue based on store address.
- Discounts sorted by customer tier (Bronze, Silver, and Gold).
- Sales split by department.
- A quick alert showing when store inventory drops below standard reorder points.

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